Google Cloud H2S

AI Builder Cup

Build in JAPAC.
Build for the world.



Team Details

a. Team name: Pilipins (Carl Vincent Kho, Raymond Borres)
b. Team leader name: Carl Vincent Kho
c. Problem Statement: Sustainability & Social Impact

Lapu-Lapu City's 129,652 households depend on four authorities for disruption notices: MECO (the power company), MCWD (the water district), the city and DepEd (the education department) for class suspensions, and PAGASA (the weather bureau). Each posts images on its own page, often late. Power and water are coupled: the water district's 147 pumps stop with the grid on 1 to 2 hours' notice. A household that

Brief about the idea

Abiso turns official outage notices into a calendar for your barangay.

Abiso is Filipino for "notice". Built first for Barangay Basak, Lapu-Lapu City, Cebu.

- Every 3 hours it reads new official posts from the offices that serve the city: MECO (the power company), MCWD (the water district), DepEd (the education department), PAGASA (the weather bureau) and the city government.
- One Gemini call per new post compiles the notice into a typed event: type, barangays, start and end, status, and a link to the source post.
- Delivered as a Google Calendar per barangay, with a "prep tonight" event the evening before.
- A web page with an ask box: a Gemini agent answers in Cebuano, Filipino or English, with source links.
- Any AI assistant can ask it through MCP (Model Context Protocol) tools and an A2A (agent-to-agent) card.

200
official posts in the live feed

392
events, refreshed every 3 hours

167/167
first batch compiled, 0 failures

$0.476
Gemini spend for that first batch

Why the notices fail

- DepEd Order 022 s. 2024: brownout and heat suspensions are the mayor's call, so they come late.
- The water district's 147 pumps stop with the grid on 1 to 2 hours' notice (SunStar Cebu, 15 Aug 2026).
- Facebook reaches 81.9% of Filipinos; a notice is an image in a feed (DataReportal 2026).
- Metro Cebu firms lose 1.4% of sales to outages; 66.4% run a generator (PIDS 2023).

Opportunities

- **How different is it from any of the other existing ideas?**
  - Existing tools each cover one kind of notice, on a site you have to visit.
  - Abiso reads every office that serves one barangay and merges them. No single office can build that view, since none owns the others' notices.
- **How will it be able to solve the problem?**
  - Each notice lands on the barangay's calendar, with a "prep tonight" event at 8 PM the evening before.
  - Anyone can ask in Cebuano, Filipino or English; every answer links the official post.
  - Advance notice removes the second condition for harm: the household that did not prepare.
- **USP of the proposed solution**
  - Compile once, then look up: one model call per post, and the facts behind every answer come from the stored events.
  - Reads the image once; serves the group chat, the calendar and other agents from one store.
  - Agent-ready: MCP tools and an A2A agent card. The offices change nothing.

Tool	Covers	Reaches you
BrownoutBA	power only	a site you check
Walang Pasok trackers	class suspensions, via news	sites, one app
Emergency Cell Broadcast	disasters only	every phone
Following each page	everything	if the feed shows it
Abiso	**power, water, classes, roads, weather**	**calendar, web, Messenger, agents**

Source: each tool's own public page. Abiso could send its power events to BrownoutBA rather than compete.

List of features offered by the solution

1 Compile once

Every 3 hours, one Gemini call (Vertex AI) per new post turns text and images into a typed, sourced event.

2 Reconcile

Merges duplicates, applies extensions and cancellations, and flags issuers that disagree (2 conflicts in the first batch).

3 Barangay calendars

One Google Calendar per barangay. A "prep tonight" event at 8 PM the evening before. Cancelled events stay, marked CANCELLED.

4 Ask box

Web page where a Gemini agent answers in Cebuano, Filipino or English, with links. With no notice it says so, with the last update time.

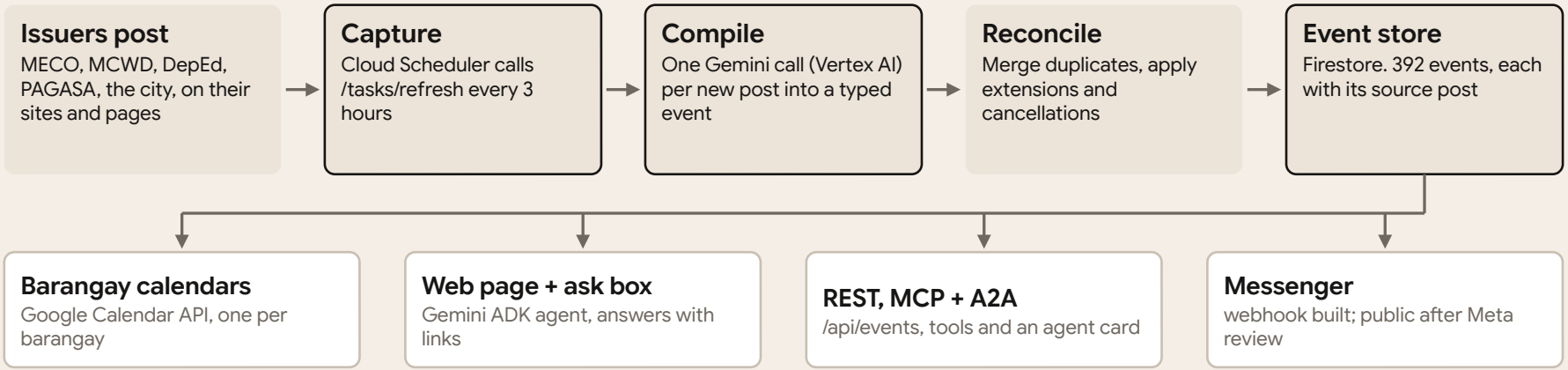
5 Agent access

REST (/api/events), MCP tools (power, water, classes, roads, weather) and an A2A agent card with five specialist skills.

6 Messenger

Webhook built: ask a question, or share a post to compile it. Public use needs Meta's review of pages_messaging.

Process flow diagram or Use-case diagram



Use case: one household in Barangay Basak

1. Subscribes once to the Basak calendar on the phone
2. MECO or MCWD posts a scheduled cut; Abiso compiles it at the next 3-hourly refresh
3. 8 PM the evening before: "Prep tonight: store water" shows on the calendar
4. Asks "Naay brownout sa Basak ugma?" and gets times plus the MECO post link

Wireframes/Mock diagrams of the proposed solution (optional)

1 Official notice (MCWD, real)

EMERGENCY

EWSI, Mactan Rock-Basak pump shut off due to MECO power interruption, 13June, 7:00 p.m. - Ongoing., AA: Sudtongan Basak, Parts of Basak proper, Marigondon, Agus, Parts of Maribago, Suba Basbas, Suba Panas, Babag 2 Gabi, Alegria, Poblacion, San Miguel, Pilipog.

EWSI = emergency water service interruption.
AA = affected areas.

2 Compiled event (real output)

```
{
 "type": "water",
 "status": "ongoing",
 "areas": [
  {"barangay": "Basak",
   "detail": "Sudtongan Basak"},
  {"barangay": "Basak",
   "detail": "Parts of Basak proper"},
  ... 10 more areas ],
 "start": "2026-06-13T19:00:00+08:00",
 "end": null,
 "source_post_ids":
  ["mcwd:2f5a5e122b91c889"],
 "model": "gemma-4-31b-it"
}
```

3 Basak calendar (mock view)

Abiso: Basak

Sat 13 Jun

7 PM No water: Basak + 6 more
ongoing, MCWD notice linked

Mon 15 Jun

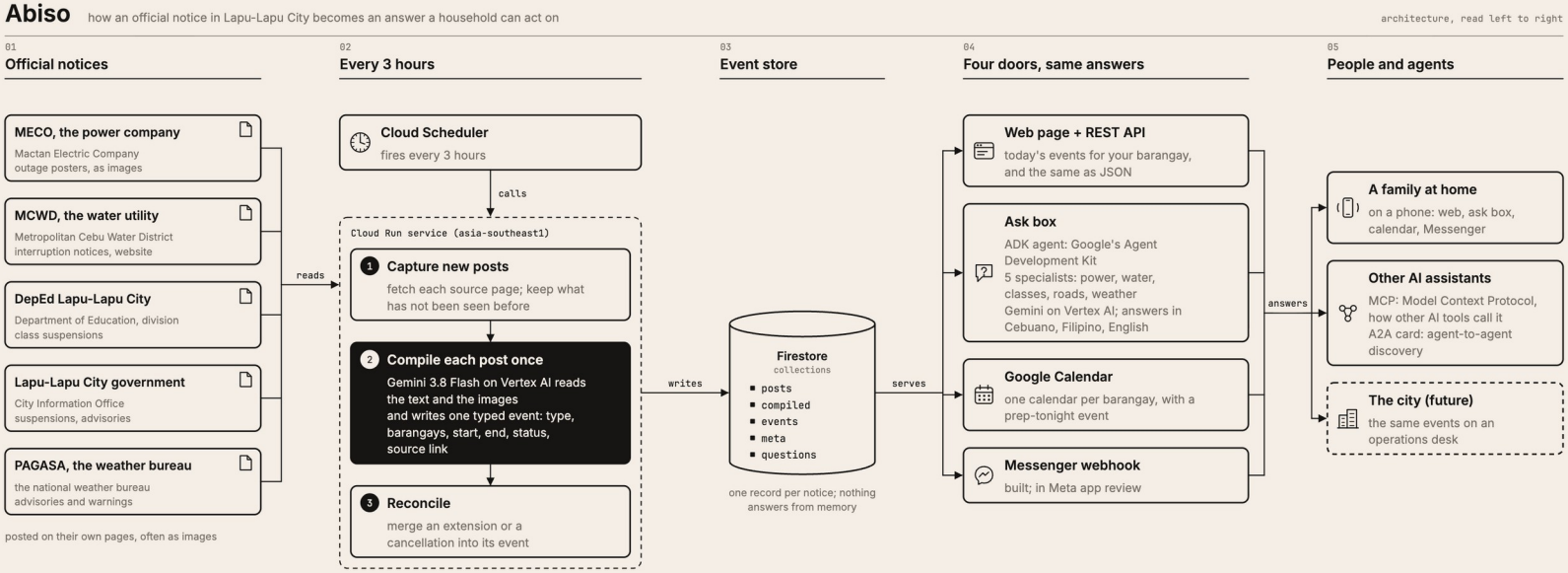
8 PM Prep tonight: store water
Water cut tomorrow 9 AM. Fill containers.

Tue 16 Jun

9 AM No water: pump repair
until Wed 9 AM, MCWD notice linked

Panels 1 and 2 are real: post mcwd:2f5a5e122b91c889 and the event compiled from it in the 2026-10-08 first batch. Panel 3 is a mock of the Basak calendar view, built from that event and the scheduled repair ev_c04fc60fd67e593c.

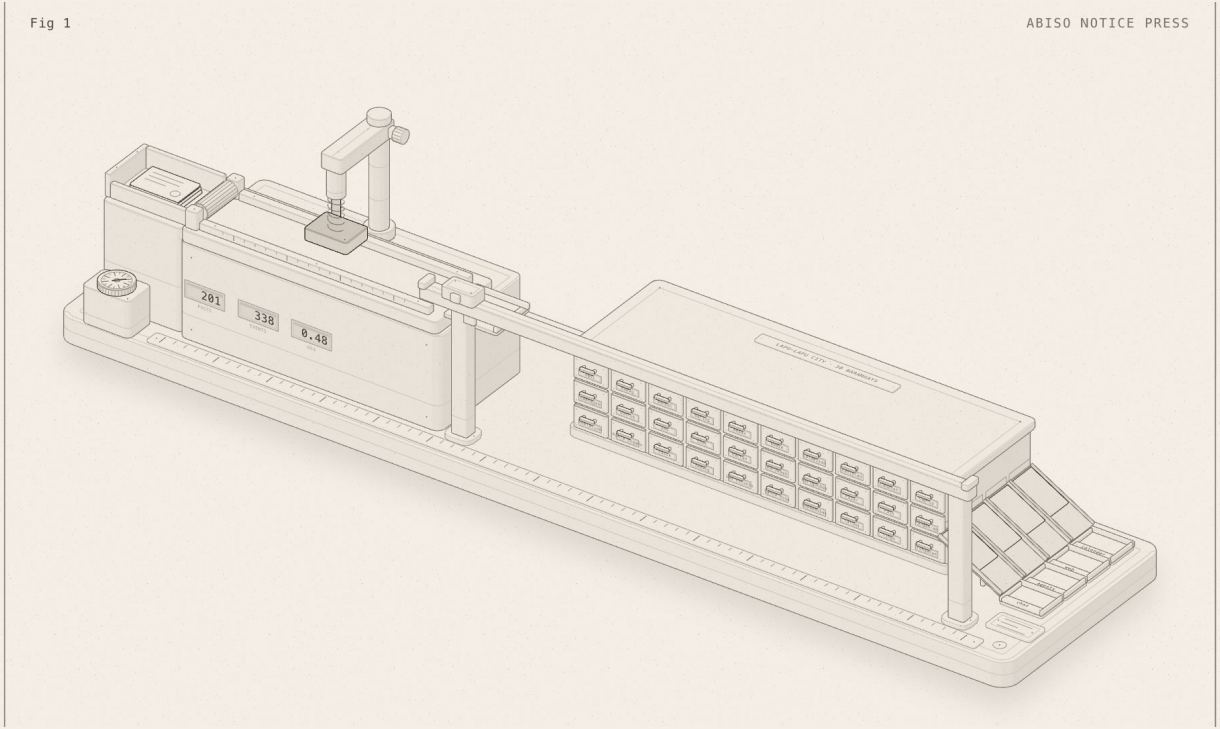
Architecture diagram of the proposed solution



SPEND CAPS one Gemini call per post (about $0.0024); each refresh at most $0.15; the ask box 8 questions per person per 10 minutes

BUILT WITH Vertex AI (Gemini 3.8 Flash), Google ADK, Cloud Run, Cloud Scheduler, Firestore, Secret Manager, Cloud Build

Architecture diagram of the proposed solution



The notice press: one stamp per post (Gemini 3.8 Flash on Vertex AI), 30 barangay drawers (Firestore), four doors (web, chat, calendar, agents). Drawn with the anatomy kit; its readout is the live feed.

Technologies to be used in the solution

Google Cloud and Google AI

Gemini 3.8 Flash on Vertex AI
compiles each new post; powers the ask-box agent. Global endpoint, service account, no API key

Google ADK
coordinator plus 5 specialist agents; A2A card via to_a2a

Cloud Run
asia-southeast1, min 1 instance during judging, max 3: page, /api, MCP, A2A, webhook, refresh

Firestore
posts, compiled results, events, feed metadata, question log

Cloud Scheduler
calls /tasks/refresh every 3 hours

Google Calendar API
one calendar per barangay, prep-tonight events

Secret Manager
holds the refresh hook's token and the Messenger keys

Cloud Build
builds the container from the repo

Artifact Registry
stores the images Cloud Run deploys

Open models and protocols

Gemma 4 31B
open model on a local GPU (vLLM): the first offline batch of 0-1 image posts, and the benchmark

MCP
the store's query functions as tools, over HTTP at /mcp/

A2A
agent card so other agents can find and ask Abiso

Python, FastAPI, Pydantic
one typed Event schema from compiler to every door

Messenger Platform
webhook for questions and shared posts

Estimated implementation cost (optional)

Measured and estimated

$0.476
Gemini spend, first batch of 167 real posts (2026-10-08)

$0.08
first scheduled-style refresh 2026-10-17: 47 posts captured, 33 new compiled

about $0.20
Gemini spend to score the 50-post test split

about $13/month
estimate: one warm Cloud Run instance during judging (list idle prices, 1 vCPU + 1 GiB)

Cost per 1,000 posts (cost study, 2026-10-04)

Compiler	USD	PHP
Gemini 3.8 Flash, default thinking	**$4.35**	**₱272**
Gemini, Batch tier	**$2.18**	**₱136**
Gemini, low thinking (accuracy not scored)	**$1.39**	**₱87**
Gemini, list price from 2027-01-01	**$8.71**	**₱545**
Gemma 4 31B on own GPU, MECO rate (electricity)	**$0.032**	**₱2.03**

Spend caps in the live service

- Each refresh capped at $0.15, all refreshes at $10 total. Ask box: 8 questions per caller per 10 minutes; Gemini stops past $2 per instance.

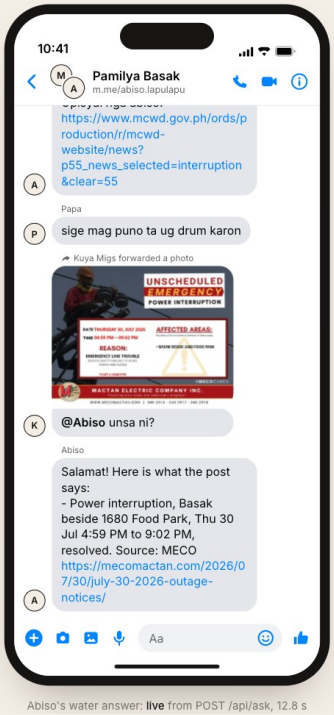
Assumptions

- Per-1,000 figures: 20 synthetic posts, an equal mix of 5 issuers (assumed), at the Gemini API list price of 2026-10-01; Vertex AI pricing not compared.
- Gemma: electricity only, hardware owned; 100 W for the rest of the PC and 90% power-supply efficiency assumed. ₱62.535 per USD (2026-10-02).
- Firestore and request charges are not measured yet.

Snapshots of the prototype



Desktop



Abiso's water answer: **live** from POST /api/ask, 12.8 s

Family group chat (mock UI, live answers)

The deployed page

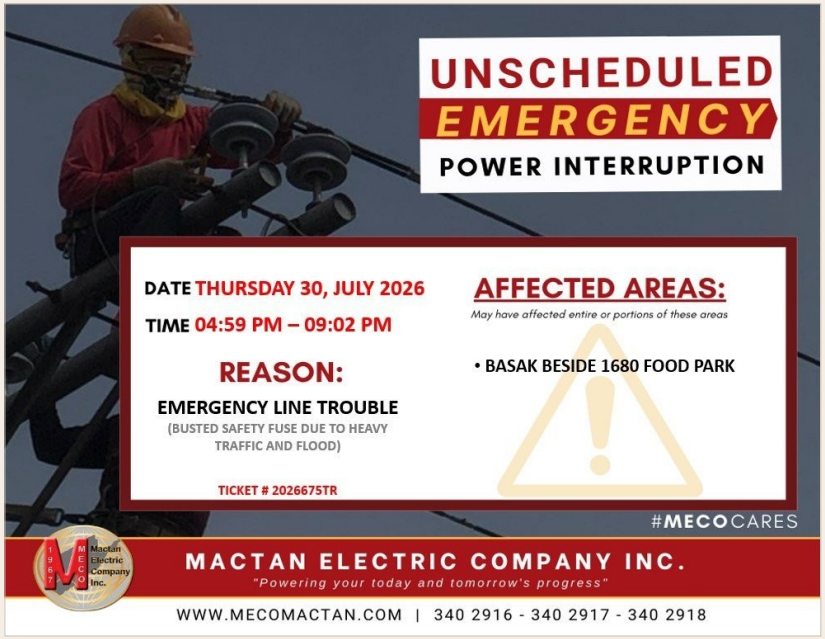
abiso-462688966054.asia-southeast1.run.app

- Header with the feed's freshness: when it last updated, how many events and posts.
- Ask box: a Gemini agent answers from the official notices only, in Cebuano, Filipino or English, and links each one.
- Notices: each card is one compiled event with its times, places, the compiling model and a link to the official post.
- Group chat: the Messenger webhook is built; the mock shows the live answer and a shared poster compiled back. Public access awaits Meta app review.

Live feed 2026-10-17: 200 posts, 392 events.

Snapshots of the prototype

Input: a real MECO post, 30 July 2026 (image 2 of 6)



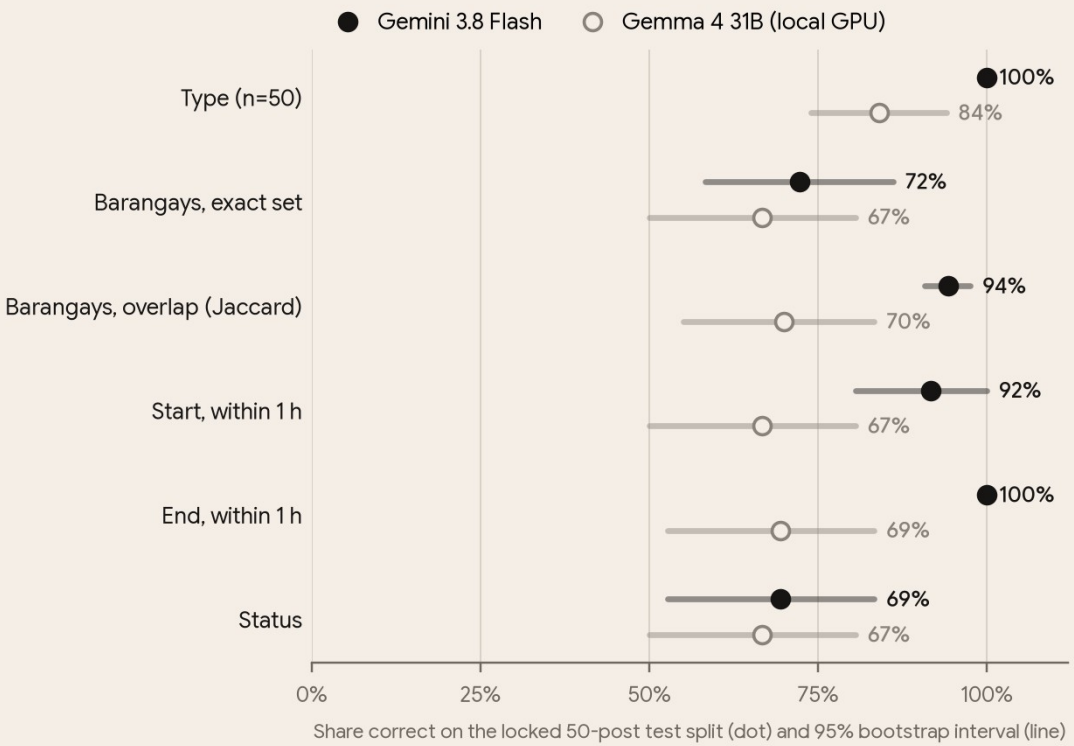
Output: the brownouts tool (MCP), same feed

```
brownouts(area="Basak",
          date="2026-07-30")

{
 "found": true,
 "events": [{
  "summary": "Emergency power
     interruption in Basak beside 1680
     Food Park on July 30 from 4:59 PM
     to 9:02 PM due to busted safety
     fuse",
  "status": "resolved",
  "start": "2026-07-30T16:59:00+08:00",
  "end": "2026-07-30T21:02:00+08:00",
  "source_links": [
   "https://mecomactan.com/2026/07/30/jul
       y-30-2026-outage-notices/"
  ]}],
 "note": "1 official notice(s) for type
   power, area Basak, date 2026-07-30."
}
```

The notice is an image with no searchable text. Abiso read it once (Gemini 3.8 Flash) and the answer comes from the stored event, with the link to MECO's post. Output captured by running the tool on the 2026-10-08 first-batch feed.

Prototype Performance report/Benchmarking



How it was measured

- Test split locked before scoring: 50 posts, SHA-256 b4d56d8f68c7.
- Labels drafted by Claude Haiku and checked by Claude Sonnet. A human spot-check of 20 found 18/20 identical.
- Type is scored on all 50; the other fields on the 36 posts labelled as disruptions.
- Gemma gave invalid JSON on 8 of 50 posts; they count as wrong.
- Paired bootstrap (10,000 resamples): Gemini is ahead on type, barangay overlap, start and end; no clear difference on exact barangays or status. The live service compiles with Gemini.
- First batch: 167/167 posts compiled, 0 failures. Live feed 2026-10-17: 200 posts, 392 events.

Status (ongoing vs scheduled vs resolved) is the weakest field for labels and models alike.

Additional Details/Future Development (if any)

1

Now
Live feed every 3 hours; Basak and city calendars

2

Next
A calendar for each of Lapu-Lapu's 30 barangays

3

City view
Disruptions per barangay per month, and where issuers disagree

4

Then
Other cities: 1,642 cities and municipalities

5

Long term
A sourced civic dataset across 42,010 barangays

Counts: Philippine Statistics Authority PSGC (30 barangays as of 2025-07-31; 1,642 and 42,010 as of 2026-06-30). Nothing here is measured yet beyond step 1.

Engineering next

- **Fine-tune Gemma 4 31B:** on the 117 training posts already split off, to close the gap to Gemini on the local GPU
- **Open Messenger:** submit pages_messaging for Meta app review; today only app testers can chat
- **Fix status:** the weakest field (69% for Gemini); add rules for ongoing vs scheduled
- **Read more pages properly:** apply for Meta's Page Public Content Access before scaling



Provide links to your:

- GitHub Public Repository: https://github.com/CarlKho-Minerva/abiso
- Demo Video Link (3 Minutes): https://abiso-462688966054.asia-southeast1.run.app/demo
- Final Product Link: https://abiso-462688966054.asia-southeast1.run.app



Traction

Five families in Lapu-Lapu City use Abiso today (interviewed offline in 2026; the four vignettes below are composites of what they told us).

The Office of the City Mayor, Lapu-Lapu City, has expressed interest through the mayor's advisors. Carl Kho demos Abiso in person to Mayor Junard "Ahong" Chan by 15 December 2026, with a memorandum of understanding as the goal.



Carl Kho with Mayor Junard "Ahong" Chan at Lapu-Lapu City Hall



An exam night and a late suspension



Ligo na mo karon: notice the night before



Power and pump out: food lost, no shower



A hot day with nothing prepared

Google Cloud H2S

AI Builder Cup

Build in JAPAC.
Build for the world.

Thank you

